

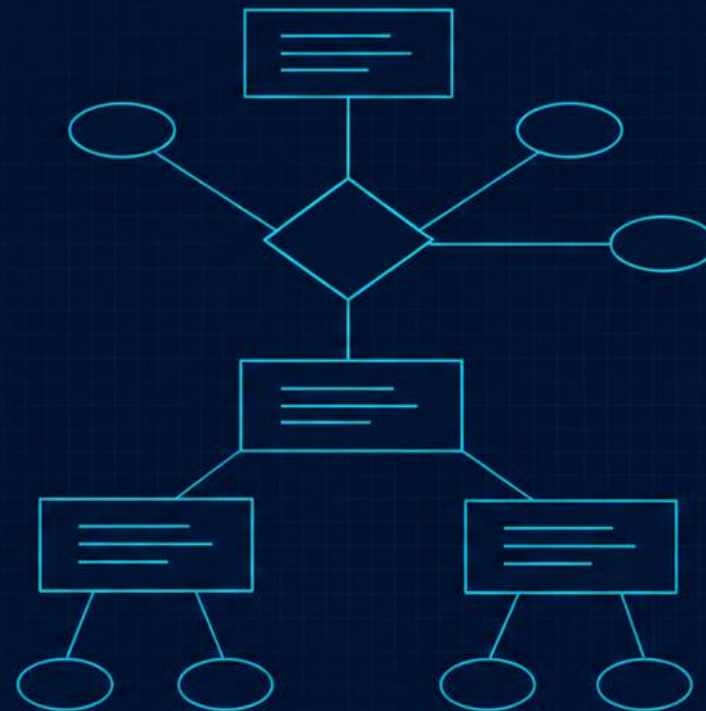


DATABASE MANAGEMENT SYSTEMS

Student And College Management System

PROJECT-BASED LEARNING

Problem Statement | Conceptual Design (ERD) | Relational Schema



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COURSE

Database
Management
Systems



CLASS

**AIML
PANTHERS**

ER DIAGRAM

The following Entity-Relationship diagram represents the core entities in the system and the relationships between them.

LEGEND



Entity

A real-world object with independent existence.



Relationship

Association between entities.



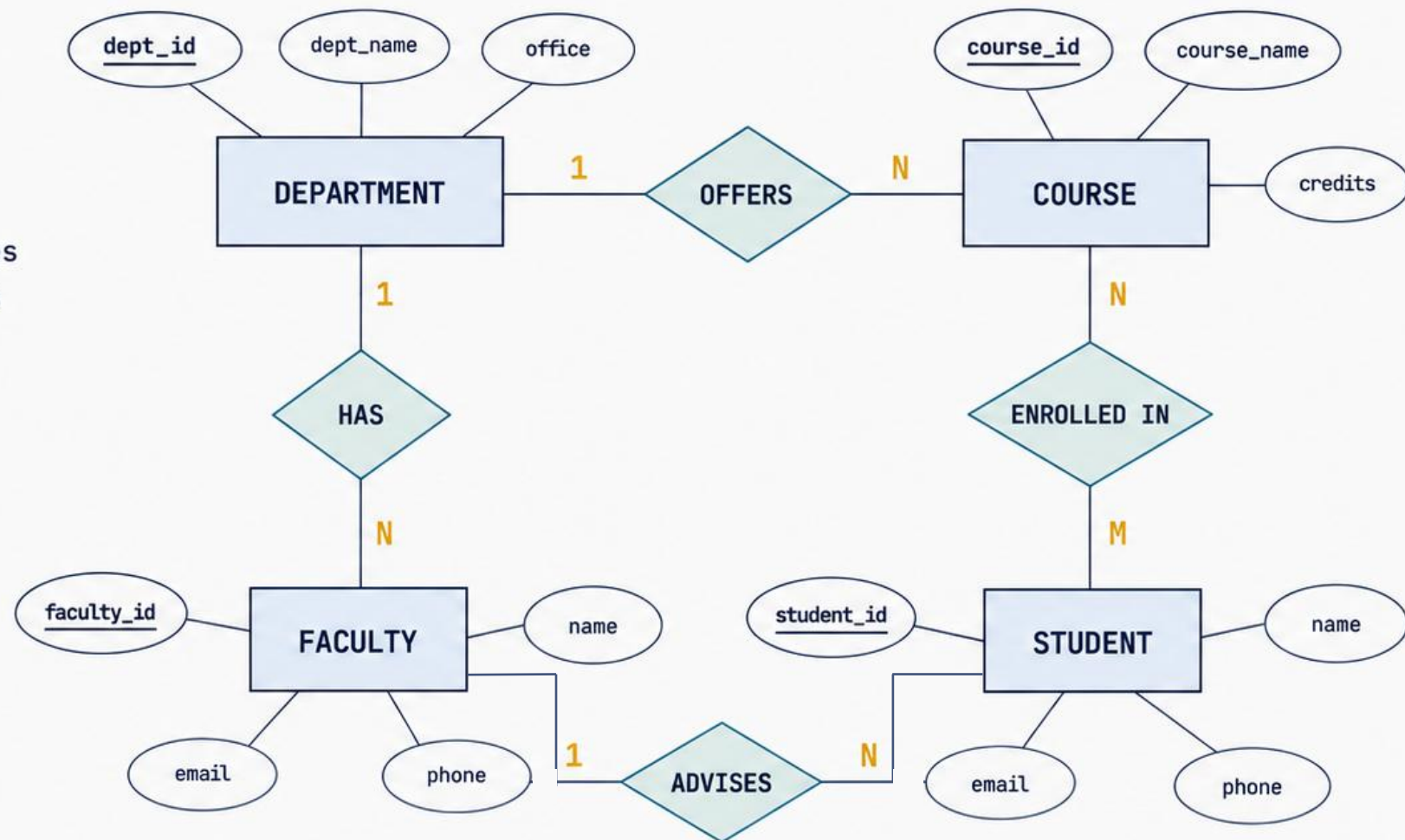
Attribute

Property or characteristic of an entity.

1:N

Cardinality

Number of occurrences in a relationship.





RELATIONAL SCHEMA

The relational schema represents the database tables, their attributes, and key constraints derived from the ER diagram.

LEGEND

PK Primary Key
Uniquely identifies each record.

FK Foreign Key
References the primary key of another table.

— Underlined Attribute
Denotes the primary key of the table.

DEPARTMENT	
Attribute	Key
<u>dept_id</u>	PK
dept_name	—
office	—

FACULTY	
Attribute	Key
<u>faculty_id</u>	PK
name	—
email	—
phone	—

COURSE	
Attribute	Key
<u>course_id</u>	PK
course_name	—
credits	—





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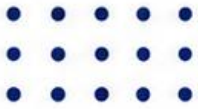
FK Foreign Key
References the primary key of another table.

— Underlined Attribute
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STUDENT	
Attribute	Key
<u>student_id</u>	PK
name	—
email	—
phone	—
advisor_id	FK (faculty_id)

TEACHES	
Attribute	Key
<u>faculty_id</u>	FK (faculty_id)
<u>course_id</u>	FK (course_id)
Primary Key: (faculty_id, course_id)	

ENROLLED_IN	
Attribute	Key
<u>student_id</u>	FK (student_id)
<u>course_id</u>	FK (course_id)
semester	—
grade	—
Primary Key: (student_id, course_id, semester)	



SAMPLE SQL QUERIES

These queries demonstrate typical operations on the database to retrieve useful information.

LEGEND



Query Purpose

What the query retrieves



SQL Query

The actual SQL statement

#	QUERY	PURPOSE
1	<pre></> SELECT * FROM STUDENT WHERE advisor_id = 101;</pre>	 List all students advised by faculty with ID 101.
2	<pre></> SELECT c.course_name, f.name FROM COURSE c JOIN TEACHES t ON c.course_id = t.course_id JOIN FACULTY f ON t.faculty_id = f.faculty_id WHERE f.name = 'Dr. Smith';</pre>	 List all courses taught by Dr. Smith.
3	<pre></> SELECT s.name, c.course_name, e.semester, e.grade FROM ENROLLED_IN e JOIN STUDENT s ON e.student_id = s.student_id JOIN COURSE c ON e.course_id = c.course_id WHERE e.grade = 'A';</pre>	 List all students who received grade 'A' in any course.
4	<pre></> SELECT d.dept_name, COUNT(c.course_id) AS total_courses FROM DEPARTMENT d JOIN COURSE c ON d.dept_id = c.dept_id GROUP BY d.dept_name;</pre>	 Show number of courses offered by each department.
5	<pre></> SELECT s.name, COUNT(e.course_id) AS total_courses FROM STUDENT s JOIN ENROLLED_IN e ON s.student_id = e.student_id GROUP BY s.name;</pre>	 List total number of courses enrolled by each student.

THANK YOU

Student and College Management System

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